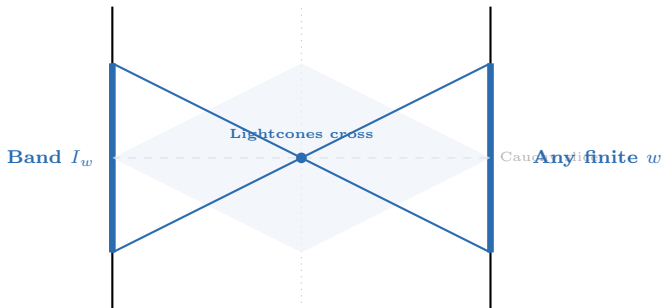
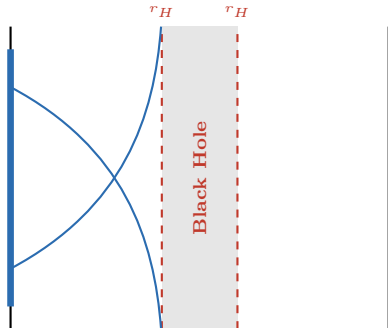


(a) Empty AdS: Causal Depth $T = \pi R$



Center $r = 0$
 For $w \geq \pi R$, domain of dependence
 covers entire Cauchy slice $\implies Y'_{I_w} = \mathbb{C}1$.
 Finite causal depth $T = \pi R$.

(b) Black Hole: Causal Depth $T = \infty$



Light rays asymptotically wrap horizon
 without crossing $\implies Y'_{I_w} \neq \mathbb{C}1$ for all $w < \infty$.
 Causal depth diverges: $T = \infty$ (Horizon signature).